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TECHNICAL REPORT

TOPIC: High Carbon Emissions

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Abstract

High carbon emissions are one of the major environmental problems contributing to climate change and its effects on the environment and human life. Carbon emissions come from many everyday activities such as transportation, electricity use, food consumption, and waste generation. Although governments and industries are taking steps to reduce emissions, individuals also need to understand how their daily activities contribute to the problem.

This report focuses on the problem of high carbon emissions and its connection with Sustainable Development Goal 13 (Climate Action). As a solution, an AI-assisted Carbon Footprint Calculator is proposed to help users estimate their personal carbon emissions and understand which activities contribute the most. Instead of only showing the total carbon footprint, the system provides personalized suggestions to help users reduce their emissions. It can also compare the user's results with average values and show the possible reduction from following different suggestions.

The proposed solution is analyzed based on sustainability principles, ISO 14001, and a simplified Life Cycle Assessment (LCA) approach. Its technical, economic, and operational feasibility is also considered. The proposed website is intended to be simple, affordable, and easy to use, while mainly focusing on creating awareness and encouraging people to make more sustainable choices in their daily lives. The solution will also serve as the basis for the website developed and demonstrated by the group in Step 2 of the project.

1. Introduction

Climate change, mainly caused by the increasing amount of greenhouse gases (GHGs) such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) in the atmosphere, has become one of the major environmental problems in the world today. Energy production, transportation, industrial activities, agriculture, and household consumption are some of the major sources of carbon emissions. While large sources of emissions such as power plants, factories, and industries are monitored and regulated, emissions are also produced through the combined activities of individuals and households. Everyday activities such as travelling, using electricity, choosing certain types of food, and generating waste all contribute to a person's overall carbon footprint.

The growth of digital technology, mobile devices, and data-based applications has created new ways to address this problem at an individual level. Information Technology (IT) tools can be used to calculate, display, and explain environmental impacts in a simple way that can be understood by the general public. A Carbon Footprint Calculator is one such tool. It is an application that estimates the amount of greenhouse gases produced by a person's daily activities and represents the result in a standard unit, usually kilograms or tonnes of CO₂-equivalent per year.

However, many existing carbon calculators available online mainly work as one-time assessment tools. They provide a final carbon footprint value but often do not help users understand what they can do to reduce it. They also provide limited options for tracking changes over time or receiving personalized suggestions. This report proposes an improved Carbon Footprint Calculator with personalized recommendations that can identify the main sources of a user's emissions and suggest practical ways to reduce them. The proposed solution is an engineering/IT-based approach to the problem of high carbon emissions and is developed in relation to SDG 13 (Climate Action).

2. Problem Statement

Global carbon emissions continue to rise despite decades of international climate negotiations and national policy commitments. A large share of this growth is attributable to individual and household consumption patterns that remain poorly understood by the very people responsible for them. The core problem addressed in this report can be summarized as follows:

- Lack of awareness: Most individuals have no quantitative understanding of how their daily transport, energy, food, and consumption choices translate into carbon emissions.
- Disconnected feedback loop: Even when awareness exists, there is no continuous, easy-to-use mechanism that links a specific lifestyle change to a measurable reduction in emissions.
- Fragmented and static tools: Existing carbon calculators are largely one-off, generic, and not localized to region-specific emission factors (for example, electricity grid emission intensity, which varies significantly by state/country).
- Absence of actionable guidance: Calculators typically stop at reporting a number; they rarely guide the user toward specific, feasible, and prioritized behavioural or purchasing changes.
- Limited engagement and retention: Without gamification or social/community elements, users rarely return to track progress, resulting in a single measurement rather than sustained behavioural change.

This report addresses the question: how can an IT-based engineering solution make personal carbon accounting accurate, engaging, and directly actionable, so that the aggregate effect of individual behavioural change becomes a meaningful contributor to national and global emission-reduction targets?

3. Sustainability Analysis

3.1 Relevant Sustainable Development Goals (SDGs)

The proposed solution is primarily aligned with SDG 13 (Climate Action), which calls for urgent action to combat climate change and its impacts, including strengthening resilience and improving education, awareness, and human/institutional capacity on climate mitigation. In addition, the solution supports:

- SDG 7 (Affordable and Clean Energy): By highlighting the carbon intensity of electricity consumption, the tool nudges users toward energy-efficient appliances and renewable energy adoption.
- SDG 11 (Sustainable Cities and Communities): Aggregated, anonymized user data can help identify high-emission activity clusters within a city, informing sustainable urban transport and infrastructure planning.
- SDG 12 (Responsible Consumption and Production): The tool's food and consumption modules encourage users to make lower-impact purchasing and dietary choices.

3.2 Alignment with ISO 14001 (Environmental Management Systems)

ISO 14001 provides a structured framework for organizations (and, by extension, individuals through this analogy) to identify, monitor, and continually improve their environmental performance using the Plan–Do–Check–Act (PDCA) cycle. The proposed Carbon Footprint Calculator operationalizes this cycle at an individual level:

- **Plan:** The user establishes a baseline footprint and target reduction goals across categories (transport, energy, food, waste).
- **Do:** The user follows personalized recommendations generated by the nudge engine (e.g., switching commute mode, reducing standby power usage).
- **Check:** The dashboard continuously tracks emissions data and compares actual performance against the baseline and target.
- **Act:** The system recalibrates recommendations based on progress, closing the feedback loop for continual improvement, consistent with ISO 14001's principle of continual environmental improvement.

3.3 Simplified Life-Cycle Assessment (LCA) Approach

A full Life-Cycle Assessment (cradle-to-grave analysis of a product or service's environmental impact) is beyond the scope of an individual-use calculator; however, the tool adopts LCA principles in a simplified form by estimating emissions across the functional stages of consumption relevant to daily life: (i) energy generation and use (electricity, cooking fuel), (ii) transportation (fuel combustion or grid electricity for EVs), (iii) food production and supply chain (embedded emissions in dietary choices, e.g., meat versus plant-based diets), and (iv) waste generation and disposal (landfill methane potential versus recycling/composting). This mirrors the Scope 1 (direct), Scope 2 (energy-indirect), and Scope 3 (value-chain indirect) emissions classification commonly used in corporate carbon accounting and adapted here for individual users.

A formal Environmental Impact Assessment (EIA) is not directly applicable to a software tool of this nature, as EIA is a regulatory requirement typically triggered by physical/infrastructure projects. However, the underlying logic of EIA — systematically assessing environmental consequences before and during an intervention — is reflected in the way the tool assesses the environmental consequence (emissions) of each lifestyle input before recommending changes.

4. Proposed Engineering/IT Solution

The proposed solution — distinct in emphasis from a conventional static calculator — is an AI-Assisted Personalized Carbon Footprint Calculator with a Behavioural Nudge Engine and Localized Carbon-Offset Marketplace. Its distinguishing features are described below.

4.1 Multi-Category Emission Input Module

Users input data across four core categories: (1) Transport — mode, distance, and frequency of daily commute (car, two-wheeler, bus, train, walking/cycling, or electric vehicle); (2) Energy — monthly electricity consumption (kWh) and cooking fuel type (LPG, PNG, electric induction); (3) Food — dietary pattern (vegan, vegetarian, mixed, high-meat) and approximate frequency of packaged/processed food consumption; (4) Waste — waste segregation practices and approximate weekly waste generated. Emission factors are drawn from recognized sources such as the IPCC emission factor database and region-specific grid emission factors (e.g., published by the Central Electricity Authority for the Indian grid), ensuring that the footprint calculation reflects local rather than generic global averages.

4.2 AI-Driven Behavioural Nudge Engine

This is the core differentiator of the proposed solution. Rather than simply displaying a total footprint, a lightweight recommendation engine (rule-based initially, extensible to a trained classification/regression model as usage data accumulates) analyses which category contributes most to a user's footprint and generates ranked, personalized suggestions — for example, recommending carpooling or public transport if commuting dominates the footprint, or suggesting specific energy-efficient appliance upgrades if electricity use is the primary contributor. Suggestions are prioritized by estimated impact-to-effort ratio, so users see the most effective, least disruptive changes first.

4.3 Gamification and Engagement Layer

To address the retention problem common to static calculators, the tool incorporates streaks (consecutive days/weeks of logging low-emission choices), achievement badges, and an optional, anonymized peer or community leaderboard. This transforms a one-time assessment into a sustained habit-formation tool.

4.4 Localized Carbon-Offset Marketplace

For emissions that cannot be immediately eliminated, the platform links users to verified, region-local offset opportunities — such as certified tree-plantation drives, community solar projects, or local NGO-run renewable energy initiatives — allowing users to voluntarily offset

a portion of their footprint while directly supporting local environmental projects rather than distant, opaque international offset schemes.

4.5 Visualization Dashboard

A responsive dashboard presents the user's footprint trend over time, a category-wise breakdown (pie/bar charts), and a comparison against city, state, national, and global per-capita averages, giving users meaningful context for interpreting their own number.

5. Feasibility Analysis

5.1 Technical Feasibility

All components of the proposed system rely on mature, well-documented, and widely available open-source technologies (React/JavaScript, Node.js/Python, Chart.js, Firebase/MySQL). Emission factor data required for the calculation logic is publicly available from sources such as the IPCC and national electricity regulators. The rule-based recommendation engine can be implemented with standard conditional logic, making the Minimum Viable Product (MVP) technically achievable within an academic project timeframe, with the ML-based enhancement positioned as a future scalability feature rather than a Phase-1 dependency.

5.2 Economic Feasibility

The solution has a low development and deployment cost. Front-end and back-end frameworks used are free and open-source; free-tier cloud hosting and database services (e.g., Firebase's free tier) are sufficient for an academic-scale deployment and initial pilot usage. No specialized hardware is required, as the tool runs entirely as a web application accessible via any standard browser.

5.3 Operational Feasibility

The tool requires only that users have periodic access to a browser and are willing to input basic lifestyle data (a low barrier to entry compared to hardware-based monitoring solutions such as smart meters). The interface is designed to be simple and self-explanatory, requiring no specialized training. Data privacy is addressed by anonymizing data used for leaderboard/community features and by storing personal activity logs securely, in line with standard data-protection practice.

5.4 SWOT Summary

Strengths	Weaknesses
Low cost, open-source stack; localized emission factors; continuous engagement via gamification	Relies on self-reported data accuracy; ML personalization requires sufficient usage data to mature
Opportunities	Threats
Integration with smart meters/IoT for automatic data capture; partnerships with local NGOs for the offset marketplace; scaling to institutional/campus-wide carbon tracking	User drop-off without sustained engagement features; competition from established global carbon-tracking apps

6. Expected Environmental Benefits

Behavioural science and pilot studies of consumer-facing carbon feedback tools consistently indicate that providing individuals with clear, personalized, and continuous feedback on resource consumption can lead to measurable reductions in energy use and associated emissions, typically in the range of five to fifteen percent, without requiring any capital investment or infrastructure change. Applied at scale, even modest per-user reductions aggregate into significant cumulative impact:

- Direct emission reduction: Encourages shifts toward lower-carbon transport modes, reduced electricity wastage, and lower-impact dietary choices.
- Indirect/awareness benefit: Builds long-term climate literacy, which influences broader consumption and voting/policy-support behaviour beyond the immediate categories tracked.
- Support for local green projects: The offset marketplace channels voluntary contributions directly into local renewable energy and afforestation projects, generating co-benefits such as improved local air quality and biodiversity.
- Data-driven policy input: Aggregated, anonymized usage patterns can help local municipal bodies identify high-emission zones or behaviours and design targeted interventions.

7. Justification of the Proposed Solution

The proposed AI-assisted calculator with a nudge engine and localized offset marketplace is justified over alternative approaches for the following reasons. Compared to purely regulatory or fiscal measures (such as carbon taxes), it operates at the individual/household level, which

is largely outside the direct reach of industrial-scale regulation, and requires no legislative change to implement. Compared to a conventional static carbon calculator, it addresses the two critical weaknesses identified in the Problem Statement — lack of continuous engagement and lack of actionable guidance — through its gamified tracking and prioritized recommendation logic. Compared to hardware-based solutions such as smart energy meters, it has a substantially lower cost and faster deployment timeline, since it requires no physical installation and can be accessed by any user with a smartphone or computer. Finally, by localizing both emission factors and offset opportunities, the tool avoids the common criticism of generic global calculators and offset schemes — namely, that they feel abstract and disconnected from the user's actual context — making sustained user engagement more likely.

8. Conclusion

High carbon emissions remain a critical global environmental challenge, and while large-scale industrial and policy interventions are essential, individual and household-level behaviour represents an underutilized lever for emission reduction. This report has proposed an AI-assisted, personalized Carbon Footprint Calculator with a behavioural nudge engine and localized carbon-offset marketplace as a practical, low-cost, and technically feasible IT solution aligned with SDG 13 and supporting SDGs 7, 11, and 12. By combining accurate, localized emission accounting with continuous engagement and actionable, prioritized recommendations, the proposed tool addresses the key shortcomings of existing static calculators and offers a scalable pathway from individual awareness to measurable behavioural change. This individual proposal will be evaluated alongside those of other group members, with one solution selected for full website implementation and demonstration in Step 2 of this project.

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